SAMUEL LYE

Junior at Cornell University with 8 months of work experience in software projects. Familiar with web and mobile technologies, web-APIs, databases and full-stack development. I am highly motivated, passionate and enjoy collaborating with other people to make things happen.

+1 (607) 310-1577 | +65 96239848 | sl2982@cornell.edu | https://samuellye98.github.io/portfolio/ | https://www.linkedin.com/in/samuellye98/

EDUCATION

BA Economics & Information Science

Cornell University

08/2018 - 05/2022 | Ithaca, NY

- Courses: Object Oriented Programming and Data Structures, Discrete Structures, Functional Programming, Web Design and Programming, Introduction to Machine Learning, Regression, Quantitative Analysis
- Activities: Cornell Venture Capital, Cornell Cup Robotics Team, Lambda Phi Epsilon International Fraternity Inc.

GPA **4.13** / 4.3

GCE A Levels

Raffles Institution

01/2015 - 11/2016 | Singapore, Singapore

- Courses: Geography, Chemistry, Economics, Mathematics, General Paper, Geopolitics, Chinese

GPA **4.0** / 4.0

TECH STACK

Backend

Docker | Express | Firebase | Flask | MongoDB | Node.js | Openshift Red Hat

Frontend

CSS3 | Expo | HTML5 | Jekyll | Javascript ES6 | ReactJS | React Native | Redux-Saga | Typescript

Languages

Python | Java | OCaml | Stata

EXPERIENCE

Software Engineer Intern

Technology and Digital Services, Ministry of Home Affairs

05/2020 - 07/2020 | Singapore, Singapore

- Research and applied state-of-the-art natural language processing technologies, in the fields of sentiment analysis, named entity recognition, relation extraction, and more, via tools like Spacy, FlairNLP, and Tensorflow.
- Built a data visualization dashboard using React, Flask, and MongoDB, to streamline data analysis for over 1000 users to quickly digest critical information and make optimal operational decisions in real time.
- Orchestrated the deployment, scaling, and management of Docker containers using Openshift Red Hat and Docker Compose, enabling 2000 words to be processed through 10 NLP models per minute.

Software Engineer Intern

Distributed Ledger Team, Govtech

05/2019 - 08/2019 | Singapore, Singapore

https://opencerts.io/ | https://www.mom.gov.sg/eservices/sgworkpass

- Full-stack development using React and Redux-Saga of an Ethereum based blockchain system, OpenCerts, which creates and verifies educational certificates, that is used by over 18 educational institutions.
- Created a digital identity verification mobile application, SGWorkPass, using the same blockchain system and React Native, that has over 1,000,000 downloads in the app store.
- Wrote unit tests and integration tests on various components to improve overall code coverage.

Battalion Human Resource and Deputy Operations Officer (Lieutenant)

Singapore Armed Forces

01/2017 - 06/2018 | Singapore, Singapore

- Spearheaded the development of a software-based attendance entry system over 6 different army bases and over 400 personnel, which led the Bafalion HR office to attain the prestigious Human Resource Audit Proficiency Award.
- Supervised HR management and an integrated supply base of over 400 personnel, 100 vehicles, and 25 military cookhouses, in a multi-component Logistics Battalion, using Excel and HR software platforms to filter, identify, and analyze data.
- Oversaw consolidation effort for mechanized and motorized vehicles, which enabled active units and training institutes to focus on operational readiness and training outcomes, while ensuring administrative excellence in the Army supply system.

PROJECTS

Cornell Cup Robotics Team

Frontend Developer & Business Team

02/2019 - Present | https://cornellcuprobotics.com

- Contributed extensively to the React frontend and Blockly interface of the web app that allows users to save and send custom scripts for an educational robot kit.
- Re-designed and developed a website for the Cornell Cup Robotics Team, primarily using HTML, CSS, JavaScript, and Jekyll.
- Planned a college-level robotics competition and worked with sponsor companies to raise awareness to around 10,000 engineering students at the top 200 schools across North America.

Cornell Venture Capital

Analyst

08/2018 - Present

- Performed industry research, investment assessments, and competitive analysis for leading VCs like Storm Ventures and Stripes.
- Assisted portfolio companies in conducting primary research and creating marketing and product improvement strategies.

Cornell Secret Santa

Frontend Developer

11/2019 - 12/2019 | https://cornell-secret-santa.web.app/

- Developed a campus-wide Secret Santa matching platform using React and Python with a group of 6 students in 2 weeks, attracting around 100 participants.

Big Red Hacks (Cornell University)

Hackathon

09/2018 - 08/2019 | https://github.com/samuellye98/corsell-app

- Created a dedicated online marketplace for Cornell students, using Jinja2, Flask, SQLAlchemy, Algolia's search API, Twilio's chat API and HERE.com's map API, with a group of 4 students in a 48 hour coding sprint.
- Presented to a panel of judges from notable tech companies and awarded winner for best use of HERE.com's API.
- Worked on a mobile version using Expo, Firebase, and React Native with working features such as logging in/out and uploading item descriptions and photos to a databse.